Dynamic Recommendation Platform with User and Merchant Agents

Semester-Wide Project

Course: Artificial Intelligence **Group 13**

Background and Motivation

Online recommendation platforms do more than predict which item a user may click next. Their recommendations influence what users discover, which products become popular, and which merchants receive attention. These interactions also change the platform itself. A positive review may increase an item's visibility, repeated recommendations may strengthen a user's existing interests, and merchants may change their prices, offers, or product descriptions in response to user feedback.

Consider a shopping platform where users repeatedly interact with recommended products. If the system focuses only on immediate clicks or purchases, it may continue showing similar popular items, reduce diversity, and provide very little exposure to new or smaller merchants. At the same time, forcing too much diversity may reduce user satisfaction. The platform must therefore balance short-term engagement with long-term satisfaction, fair merchant exposure, recommendation diversity, and business objectives.

In this project, you have to develop a **Dynamic Recommendation Platform with User and Merchant Agents**. User agents should browse, like, review, ignore, and purchase items, while merchant agents may respond to reviews, change offers, update product information, or introduce new items. The recommendation platform should learn from these evolving interactions and produce recommendations that remain useful, diverse, fair, and personalised over time.

The project is motivated by *RecInter*, which introduces an interaction-centred simulation platform where user actions dynamically change item attributes and merchant agents respond to those actions (Jin et al. 2025). It is also related to the *Tunable LLM-based Proactive Recommendation Agent*, which formulates recommendation as a multi-step process that adapts to real-time feedback and optimises long-term rewards (Wang et al. 2025). Recent work on long-term LLM-based recommendation further shows the importance of learning from indirect feedback, such as continued engagement or silent disengagement, rather than relying only on explicit ratings (Zhang et al. 2025).

Module 1: Search and Constraint Satisfaction

You have to represent users, items, merchants, categories, reviews, and interactions using user–item and merchant–item graphs. Search algorithms should be used to retrieve relevant products, discover related categories, identify alternative items, and find recommendation paths from a user's known interests to new but potentially relevant products. You also have to formulate recommendation generation as a Constraint Satisfaction Problem. The constraints may include the user's budget, product availability, category preferences, minimum diversity, merchant-exposure requirements, delivery restrictions, and limits on repeatedly recommending similar items. The system should distinguish between mandatory user constraints and platform objectives such as diversity and fairness, and it should explain when all objectives cannot be satisfied simultaneously.

Module 2: Recommendation Planning

You have to model recommendation as a multi-step planning problem rather than producing an independent list at every interaction. The available actions may include recommending a familiar item, introducing a new category, asking a preference question, comparing products, offering a lower-cost alternative, presenting a merchant promotion, or delaying a recommendation until more information is available. Each action should have suitable preconditions and expected effects on user knowledge, interest, satisfaction, and future behaviour. The planner should construct recommendation campaigns that gradually introduce new products without ignoring the user's existing preferences. It should also revise the plan when the user rejects an item, changes the budget, provides a negative review, or shows interest in a previously unexplored category.

Module 3: Reasoning under Uncertainty

You have to develop a probabilistic model for estimating user preferences, purchase probability, merchant reliability, and the expected effect of a recommendation. The system should account for uncertainty because a click may indicate curiosity rather than preference, a purchase may be made as a gift, and the absence of feedback may represent either dissatisfaction or lack of attention. Bayesian reasoning or another suitable probabilistic method should be used to update beliefs as new interactions occur. The system should also estimate whether reviews and merchant-provided information are reliable and should avoid becoming overly confident based on a small number of user actions.

Module 4: Contextual Bandits or Reinforcement Learning

You have to formulate long-term recommendation as a contextual-bandit or Reinforcement Learning problem. At each interaction, the system may recommend a popular item, a highly personalised item, a diverse alternative, a product from a less-exposed merchant, or an exploratory item from a new category. The state should represent the user's evolving preferences, recent interactions, satisfaction, budget, recommendation history, item popularity, and merchant exposure. The reward function may consider clicks, purchases, ratings, continued engagement, recommendation diversity, long-term satisfaction, merchant fairness, and penalties for repetitive or irrelevant suggestions.

You may implement a contextual bandit, Q-learning, actor–critic, PPO, or another suitable learning method. The learned policy should be compared with fixed approaches such as recommending only the most popular items, selecting items with the highest predicted purchase probability, or applying a fixed diversity rule. The evaluation should examine whether short-term engagement improvements create negative long-term effects such as echo chambers, popularity concentration, or user fatigue.

Module 5: Generative AI

You have to develop a multi-agent GenAI environment containing user, merchant, recommender, fairness, and platform-coordinator agents. User agents should have different interests, budgets, decision styles, and willingness to explore unfamiliar products. Merchant agents should respond to reviews, modify offers, update product information, and attempt to improve the visibility of their items. The fairness agent should monitor whether particular merchants, product categories, or user groups receive systematically unequal treatment.

The recommender agent should understand natural-language preferences, ask clarification questions, explain recommendations, and revise suggestions after receiving feedback. However, the LLM should use the outputs of the search, CSP, uncertainty, and reinforcement learning modules instead of independently selecting products. The final explanation should describe why an item was recommended, how it matches the user's preferences, whether it was introduced for diversity or exploration, and what trade-offs were considered.

Expected Outcome

The final system should demonstrate how interactions among users, merchants, and the recommendation platform create an evolving recommendation ecosystem. You should evaluate the system over multiple rounds rather than using only independent recommendation requests. The scenarios should include changing user interests, new merchants, price changes, negative reviews, merchant responses, popularity growth, limited budgets, and users who become dissatisfied with repetitive recommendations.

A successful implementation should maintain useful recommendations while improving diversity, long-term satisfaction, and fair merchant exposure. The evaluation may consider click and purchase rates, cumulative reward, user retention, recommendation diversity, novelty, merchant-exposure fairness, popularity concentration, preference-prediction accuracy, user satisfaction, and the platform's ability to adapt to changing behaviour.

References

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